

ISO PAS 8329 (xMCF)

Status, Collaborative Use, Next steps

91st Plenary ISO TC184/SC 4/ WG 12 Meeting 90th
Plenary

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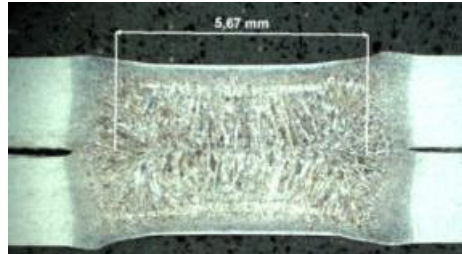
Topics for today

- ISO PAS 8329 (χ MCF) status and current activities
- PDM-IF approach for the federative use of χ MCF with AP242 for fasteners
- How to proceed with ISO PAS 8329?

Joining
technologies
& processes
are challenging ...

Challenges wrt. Connection Information: Big Variety & Complexity

- > 60 known connection techniques.
- Up to 25 quality criteria per connection technique.



Spot Weld (Section)

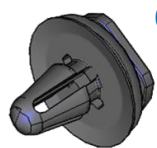
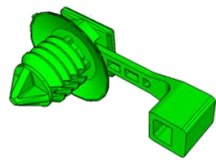
All kinds of screws



Rivets



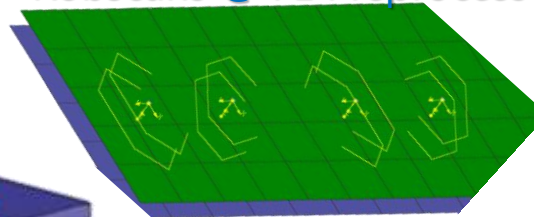
Nails



Clips



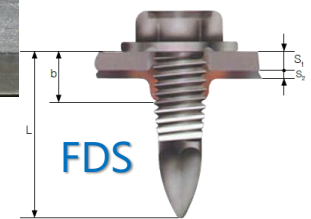
Robscans @ FE Preprocessor



Clinch Rivet Stud



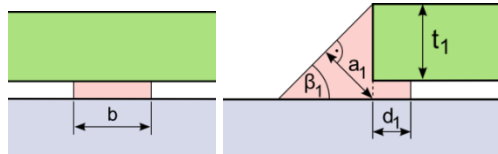
ROTAV



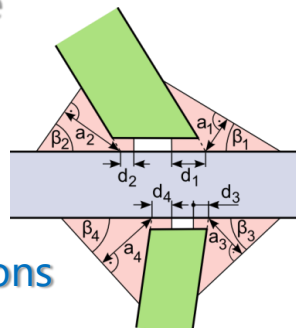
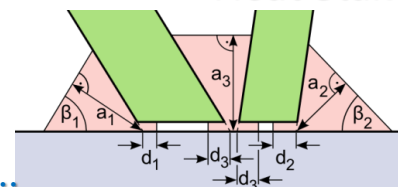
FDS



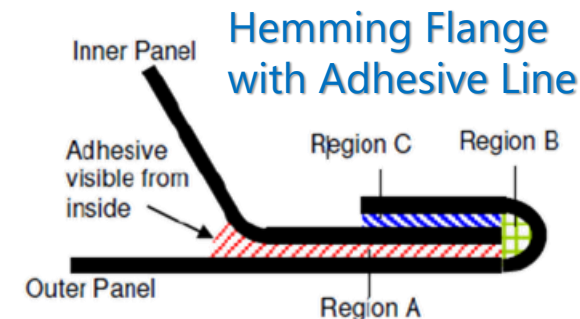
Clinch



Heat Stake



Weld lines with increasingly complex cross sections

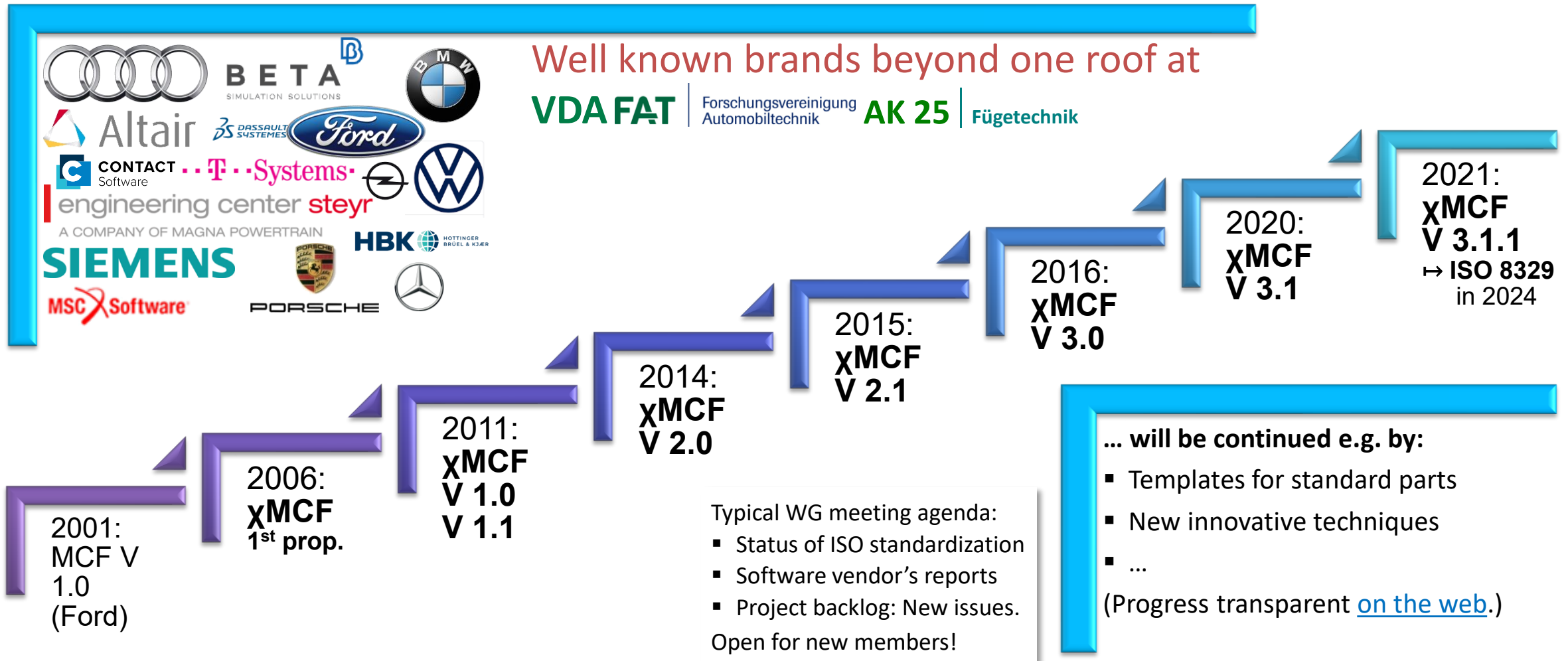


Adhesive Face

What are the Frequent Problems?

- Every OEM creates *own macros specific to each CAD system* or buys *proprietary software*.
 - Common suppliers *need to be familiar with all those tools*.
 - Data exchange along process chain needs *additional tools*, frequently “home-brewed”.
⇒ Expensive and error-prone.
 - However, in reality only *few techniques* are supported with only a *fraction of their data*.
 - Inventing new techniques or adding new parameters results in *excessive costs* and *process threats*.
 - Changing software vendors implies *high investments*.
- ⇒ Resulting “*vendor lock-in-effect*” impedes competition and hence *hinders progress*.
- ✓ xMCF is the remedy for all these issues.

χMCF – A Standard with History and Broad Support



New participants since Autumn 2025

- AFNeT Services
- Altair Engineering Inc.
- Arup
- CoreTechnologie Group
- GNS mbH
- PEM – PennEngineering
- PTC
- Robert Bosch GmbH
- The Boeing Company

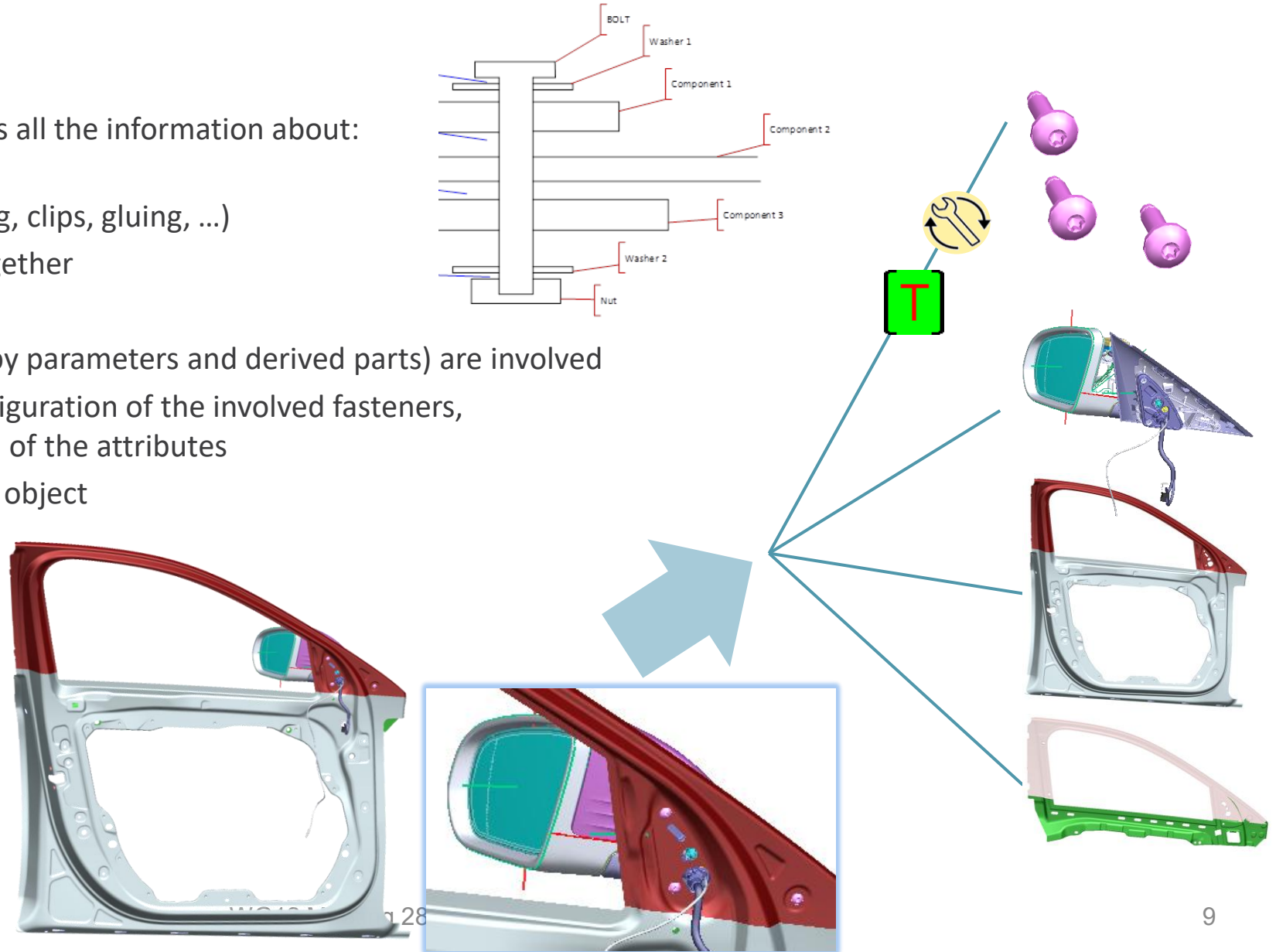
Current Activities since Autumn 2025

- The working group is preparing an update to the standard.
- In future, support will be provided for repeated connectors and connectors from catalogs.
- The verification of χ MCF files will no longer rely solely on XML schemas. Additionally, Schematron — a tool familiar from STEP — will be used. It allows for significantly more powerful checks.

PDM-IF Scope

Fasteners

- Mating bundle object, that holds all the information about:
 - Type(s) of mating (screwing, welding, riveting, clips, gluing, ...)
 - Which parts are mated together
 - In which order (stacking)
 - Which fasteners (defined by parameters and derived parts) are involved
 - In case of 150% BoM: configuration of the involved fasteners, of their 3D positioning and of the attributes
- Having a stable ID of the mating object in case of geometrical changes
- In scope of CAx-IF:
geometric elements (like contact faces)

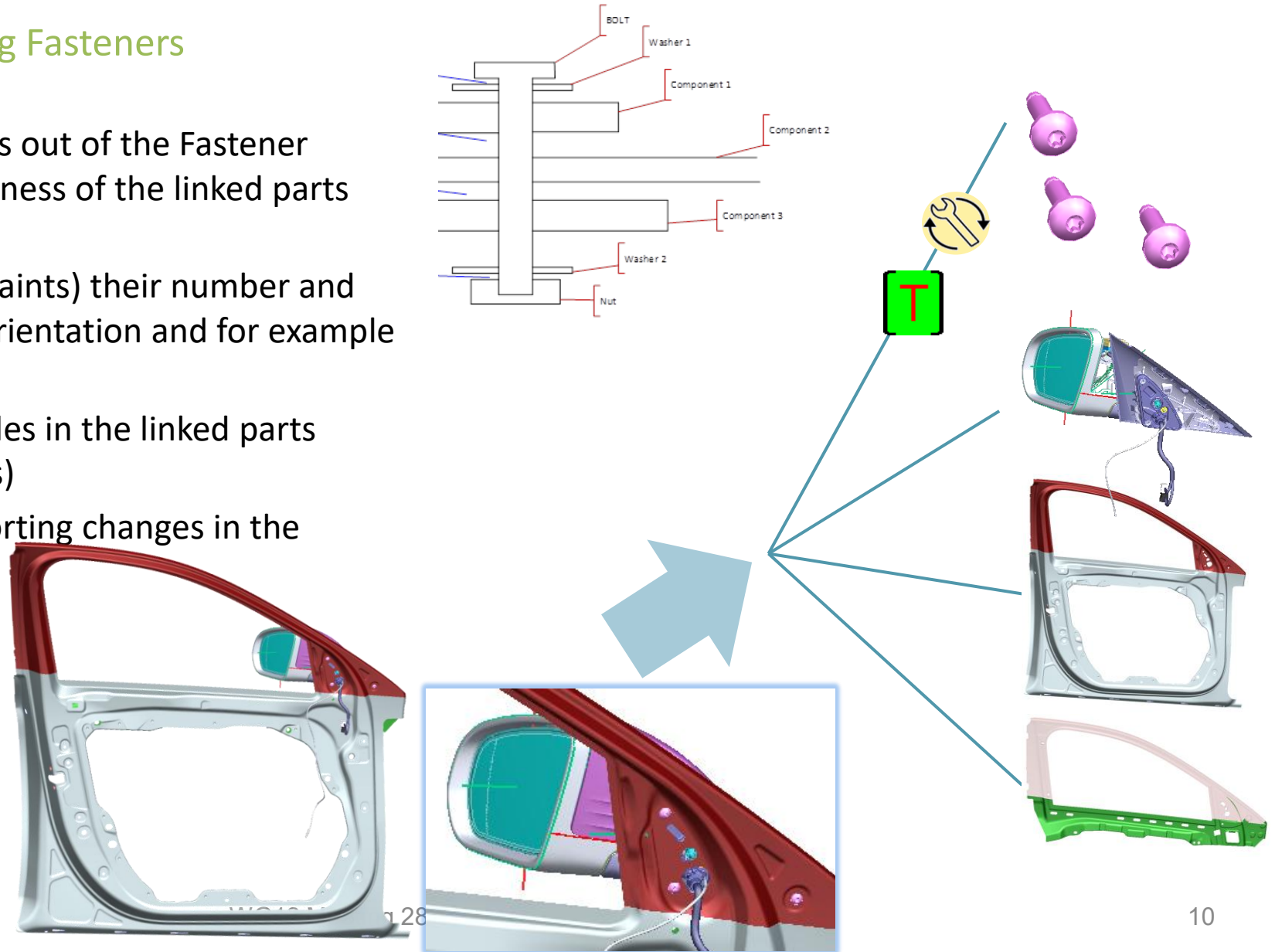


PDM-IF Scope

Modern PLM Features regarding Fasteners

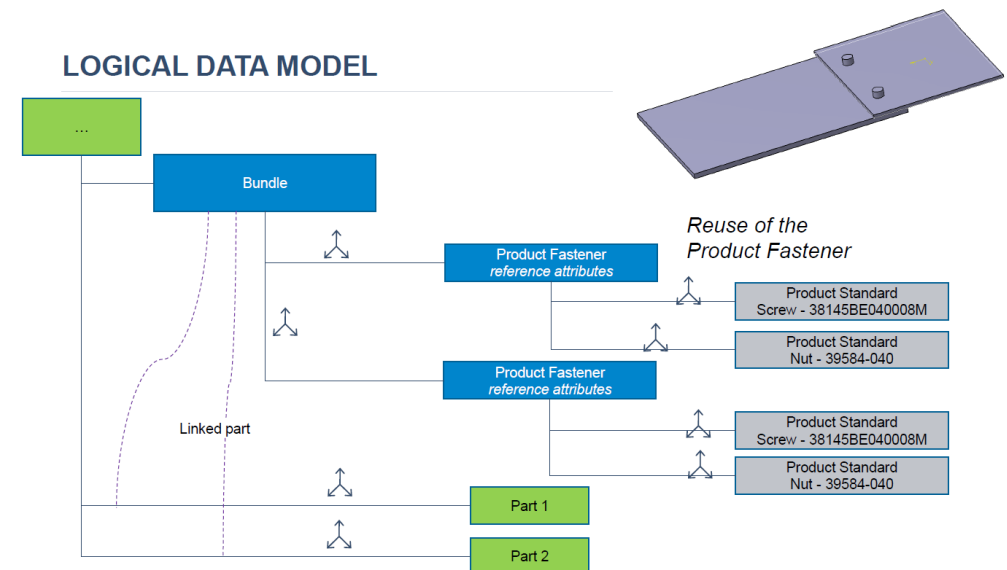
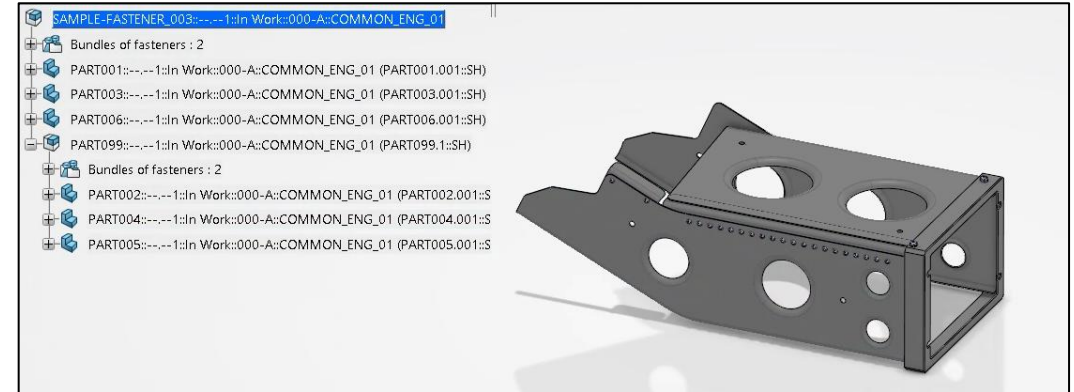
- Propose concrete fastener parts out of the Fastener parametric definition, the thickness of the linked parts and the material constraints
- Compute (from material constraints) their number and positioning (from the wished orientation and for example equidistant)
- Parametric definition of the holes in the linked parts (no physical design of the holes)

⇒ For an automated design, supporting changes in the product design with less effort.



Fasteners@Dassault Aviation

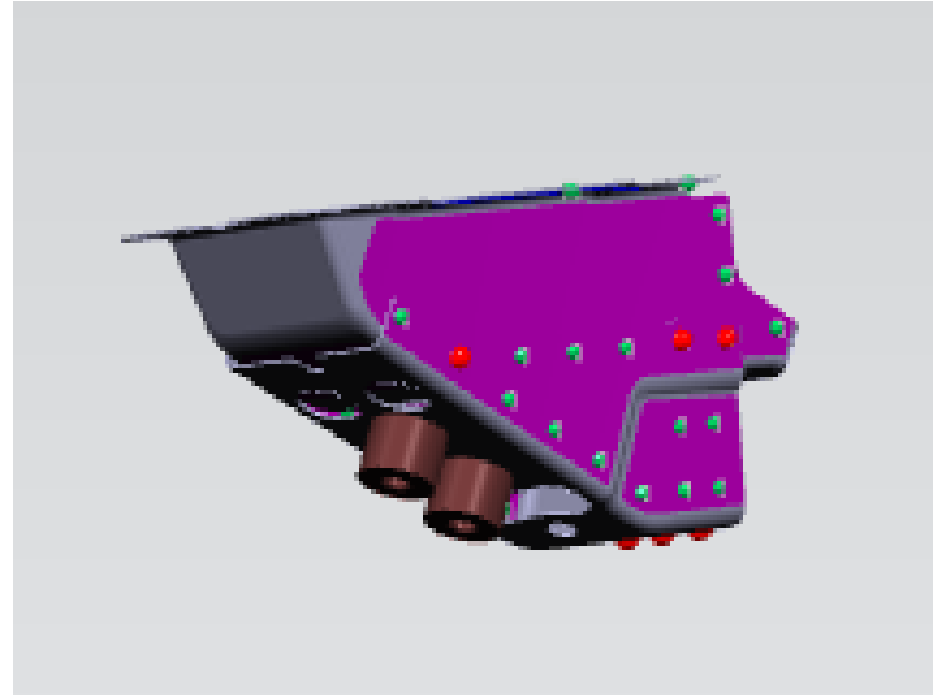
- Full dataset ready for Dassault Aviation Use Case
 - Assembly Structure mapped from 3DExp
→ CATIA V5 → AP242 XML Edition 4 + P21/JT
 - Enriched with Bundle definition, references to χ MCF file and OccurrenceRelationships between LinkedParts and ProductFasteners (parametric definition of the holes)
 - χ MCF file containing the structure and the reference/instance attributes of the Bundles and the ProductFasteners
 - Mapping of the attributes to standard attributes or customized attributes
 - Beta CAE could read this dataset into ANSA



Use Case Status

Fasteners@BMW

- Full dataset ready for BMW Use Case
- χMCF File provided by BMW
 - containing the structure and the reference/instance attributes of Carrier Sheets and Joining Elements
 - Many customized attributes seem to contain PDM data
- Assembly Structure (stpx) + Geometry (JT)
 - Enriched with Fasteners and reference to χMCF file
 - Beta CAE could read this dataset into ANSA
- Analysis of the used χMCF custom_attributes
 - Proposal where to move some of them to AP242 XML file and complete the dataset with combination of AP242 and χMCF



Use Case Status

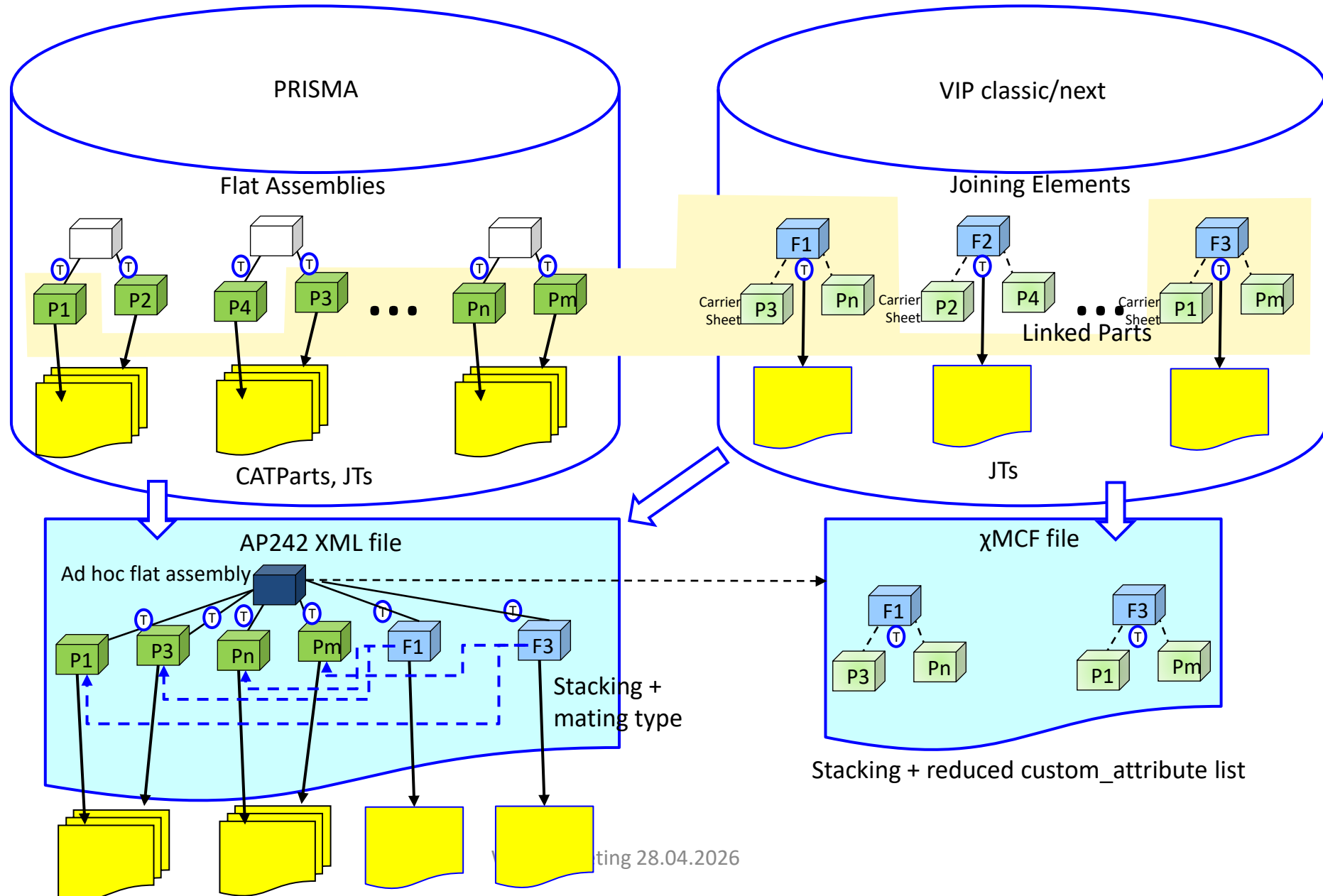
Fasteners@BMW

Computing of
Bounding Box
according to
the $\odot$ s

Extraction of the Parts
and Joining Elements

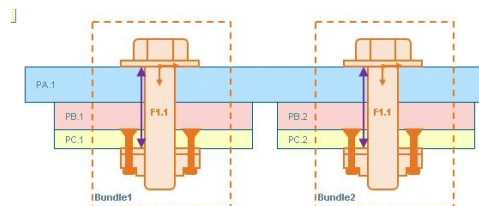
Merging VIP results
into Product Structure

- minimum use of χ MCF custom_attributes
- maximum use of AP242 attributes
- identical key attributes (label + instance of the parts/joining elements) in both files

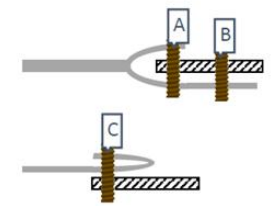


Fastener Issues

- Issues with AP242 Edition 4
 - Extension of the data model needed to support reuse and techniques typical of the aerospace industry
 - χ MCF ISO PAS Standard covering the definition of Fasteners and supported by multiple (simulation) tools
 - ⇒ decision to combine AP242 XML (product structure + basic mating capabilities) and χ MCF (high level of details of the connection)
 - ⇒ χ MCF XML file gets referenced by the AP242 XML file
- Issues with χ MCF
 - Extension of the data model needed to support reuse and techniques typical for the aerospace industry
- Requirements to both data models
 - Add to AP242 the bundle of multiple fasteners (of potentially different types) for one set of mated parts
 - Enable multiple reuse of the same fastener (defined only once)
 - Add the PLM Ids to χ MCF for smooth cross references between AP242 XML and χ MCF XML
 - Add to AP242 the support of different stacking orders for the fasteners of the same bundle
 - Add hierarchy of fasteners



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Open Questions

- However, it has not yet been decided
 - whether the update to ISO 8329 will be released as “PAS” again
 - or whether we will make the switch to “IS”.
- Furthermore, it is unclear
 - whether the standard can continue to be maintained as a Word file (with examples and test cases as attachments)
 - or whether we need to switch to the OSD format.

We need recommendations from SC4 for this.